

MODULE: V

General Structure of Electrical Power Systems:

An electric power system is a network of electrical components that supply, transfer, and use electric power. The general structure of an electric power system can be divided into three main components:

1. Generation- Source of Energy
2. Transmission-Transmission of Energy over long distance (close to the speed of light)
3. Distribution- Consumption of Energy

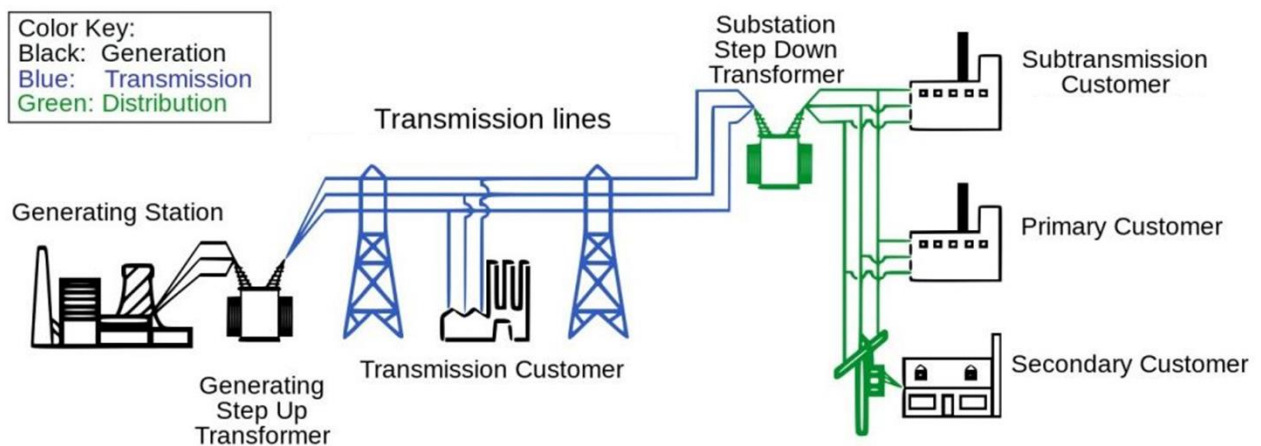


Fig. General Structure of Electrical Power System

GENERATION SYSTEM:

In generating station, the fuel (coal, water, nuclear energy, etc.) is converted into electrical energy. The electrical power is generated in the range of 11kV to 25kV, which is step-up for long distance transmission. **The power plant of the generating substation is mainly classified into three types, i.e., thermal power plant, hydropower plant and nuclear power plant.** The generator and the transformer are the main components of the generating station. The generator converts the mechanical energy into electrical energy. The mechanical energy comes from the burning of coal, gas and nuclear fuel, gas turbines, or occasionally the internal combustion engine. The transformer transfers the power with very high efficiency from one level to another. The power transfer from the secondary is approximately equal to the primary except for losses in the transformer. The step- up transformer will reduce losses in the line which makes the transmission of power over long distances.

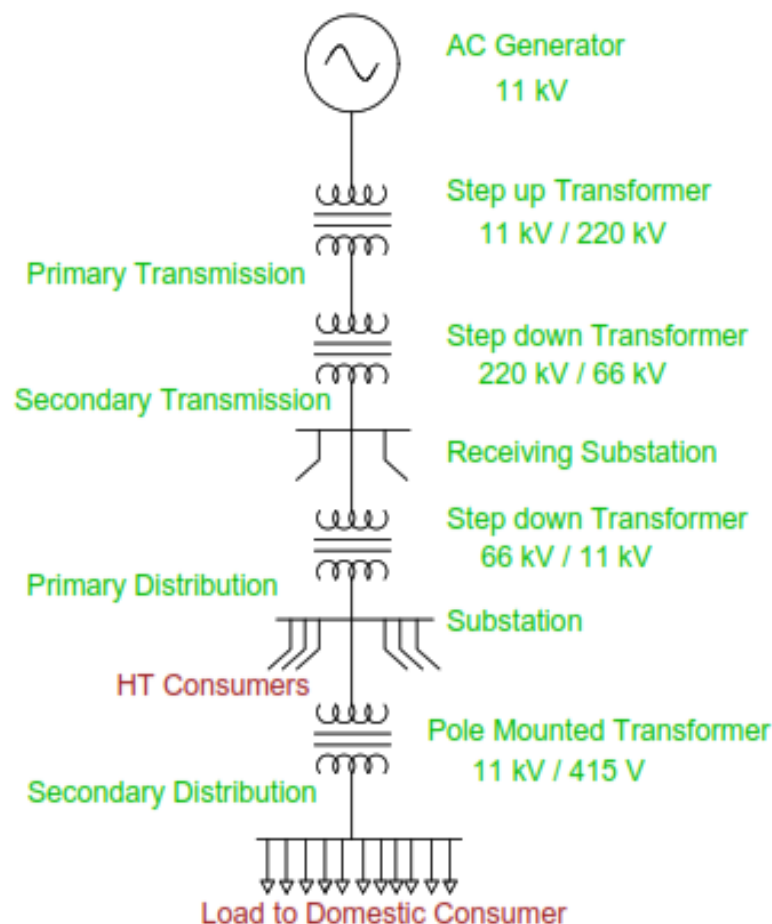


FIG : Single Line Diagram of Power Supply System

TRANSMISSION SYSTEM:

Electrical power transmission involves the bulk movement of electrical energy from a generating site, such as a power station or power plant, to an electrical substation where voltage is transformed and distributed to consumers or other substations. The interconnected lines that enable the movement of electrical energy are known as a “transmission network,” and these form an electrical power transmission system—or, as it is more commonly known, the power grid.

Primary transmission

When it is generated at a power station, electrical energy will typically be anywhere between 11kV and 33kV. Before it is sent to distribution centers via transmission lines, it is stepped up using a transformer to a voltage level that can be anywhere between 100kV and 700kV or more, depending on the distance that it needs to be transmitted; the longer the distance, the higher the voltage level.

The reason electrical power is stepped up to these voltage levels is to make it more efficient by reducing the I^2R losses that take place when power is transmitted. When voltage is stepped up, the current reduces relative to the voltage so that power remains constant, thus reducing these I^2R losses. This stage is known as primary transmission—the transfer of a large quantity of electrical power from the initial generating station to the substation via overhead electrical lines. In some countries, underground cables are also used in cases where transmission takes place over a shorter distance.

Secondary transmission

When electrical power reaches a receiving station, the voltage is stepped back down to a voltage typically between 33kV and 66kV. It is then sent to transmission lines emerging from this receiving station to electrical substations closer to “load centers” such as cities, villages, and urban areas. This process is known as secondary transmission. When electrical power reaches a substation, it is stepped down once more by a step-down transformer to voltages closer to what it was generated at—usually around 11kV. From here, the transmission phase graduates to the distribution phase, and electrical power is used to meet demand from primary and secondary consumers.

DISTRIBUTION SYSTEM:

That part of power system which distributes electric power for local use is known as distribution system. In general, the distribution system is the electrical system between the substation fed by the transmission system and the consumers meters, consists of:

(i) Feeders: A feeder is a conductor which connects the sub-station (or localized generating station) to the area where power is to be distributed. Generally, no tapping are taken from the feeder so that current in it remain the same throughout. The main consideration in the design of a feeder is the current carrying capacity.

(ii) Distributor: A distributor is a conductor from which tapping are taken for supply to the consumers.

(iii) Service mains: A service main is generally a small cable which connects the distributor to the consumers’ terminals.

Power system is divided into three parts; power generation, transmission, and distribution. In power generation, one form of energy gets converted into electrical energy. We produce electrical energy from various natural sources.

These sources are classified into two types renewable sources and non-renewable sources. In a present power system, most of the electrical energy gets generated from non-renewable sources like coal, oil, and natural gases. But these sources are limitedly available. So, we have to use these sources carefully and always find an alternate source or move on to renewable sources.

The renewable sources include solar, wind, water, tidal, and biomass. These sources are the environment-friendly, free and infinite resources available.

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